政池 明

No. 1

Date 94 · 10 · 14

§ Historical Introduction

§ Invariance

§ Why can we blieve in quarks.

- ° strong Interaction
- ° Electro-magnetic Interaction and Parton

§ Week interactions and Neutrino

- ° Perkins, Introduction to High Energy Physics (addison-Wesley Pub. Co.) 1982
- ° F. Halzen & A.D. Martin Quarks and Leptons - an Introductory Course in Modern Particle Physics

光 $E = h\nu$ $p = \frac{h}{\lambda}$ $1eV = 1.6\times10^{-19}J$

物質波 de Broglie $\lambda = 1.2\times10^{-6}\,m$

素粒子 1 GeV - 10 TeV

1911 Rutherford Scattering α線 larg angle scattering

1932 Chadwick 中性子

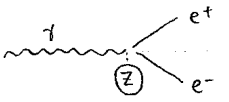
1932 Cockcroft-Walton 加速器 $p + {}^7Li \rightarrow 2\,{}^4He$

1934 荒勝文策

1930 Cyclotron Lawrence

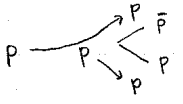
1950 Synchrotron  magnet

素粒子を人工的に作る

electron  pair creation

$0.5\,MeV/c^2$

meson $p \rightarrow p \rightarrow \pi$ $140\,MeV/c^2$

anti-proton  $p + p \rightarrow 3p + \bar{p}$ ターゲットに核子を用いた ①

$p, \bar{p}$ $938\,MeV/c^2$ Bevatron

相対論効果

Lab $\overset{m}{\circ} \longrightarrow \overset{m}{\circ}$ E_{lab} 全エネルギー $E_{lab} \sim \frac{E_{cm}^2}{2m}$ ②

CM $\overset{m}{\circ} \rightarrow \leftarrow \overset{m}{\circ}$ E_{cm}

$E^2 = m^2c^4 + p^2c^2$

$E_{lab} \sim 400\,GeV \rightarrow E_{cm} \sim 28\,GeV$

$E_{lab} \sim 5\,TeV = \underbrace{50\times50}\,GeV$

$E_{cm} = 100\,GeV$

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物質 quark

Q/|e|				
$+\frac{2}{3}$	u	c	t	···
$-\frac{1}{3}$	d	s	b	···

1950 s 100 ~ 300 MeV/c²

1974 c ~ 1.5 GeV/c² $c\bar{c}$

1977 b 4.1 ~ 4.5 GeV/c² $b\bar{b}$

(1994 t 174 GeV/c² $t\bar{t}$)

p = uud $\pi^+ = u\bar{d}$

n = udd $K^0 = \bar{s}d$

Λ = uds $J/\psi = c\bar{c}$

Ω = sss

Pauli → color (Y. B. G)

J/ψ BNL

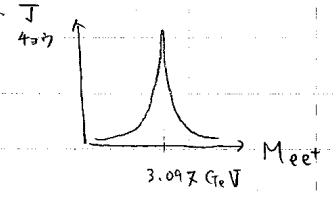
$p \longrightarrow p$ (28 GeV) $\frac{\partial\theta}{\partial\theta} = 14°$ → e^+, e^-

3.097 GeV/c²

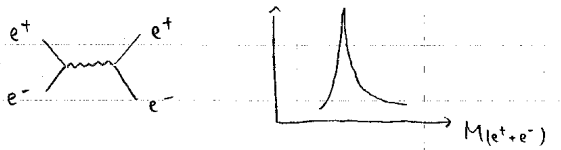
結果に従って $\theta = 14°$ (3)

S. Tung

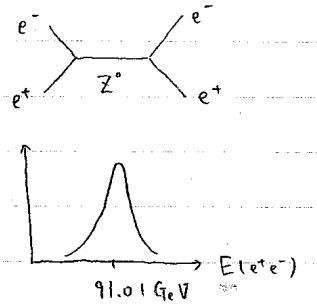
但し $E_p \gg E_e$



SLAC $e^+ \rightarrow \leftarrow e$



1989 LEP



1977 Lederman $\Upsilon : b\bar{b}$

$p \longrightarrow p$ → μ^-, μ^+

1987 Tristan 30 GeV × 30 GeV

e^+ e^-

1994 Fermi lab

$p\bar{p} \rightarrow t\bar{t}$

$t \rightarrow W^+ + b$

→ $e^+\nu$ or $\mu^+\nu$

→ $u\bar{d}$ or $c\bar{s}$

Leptons

↕ charge 0, ±e

Anti-leptons

0.51 105 1870 MeV/c²

$\begin{pmatrix} e^- \\ \nu_e \end{pmatrix} \begin{pmatrix} \mu^- \\ \nu_\mu \end{pmatrix} \begin{pmatrix} \tau^- \\ \nu_\tau \end{pmatrix}$

▷ Oct 21

ν : longitudinally polarized

$J_z = -\frac{1}{2}$ (left-handed)

z : velocity vector 方向

$\bar{\nu}$: $J_z = +\frac{1}{2}$ (right-handed)

mass が有限で left, right は従う

lepton # conserve for each type

$\gamma \rightarrow e^+ e^-$

L_e 0 −1 +1

$\pi^+ \rightarrow \mu^+ \nu_\mu$

L_μ 0 −1 +1

1983 $W^\pm$ Z^0 の発見 (CERN)

EM-int と Week-int の統一

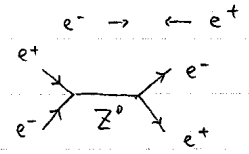
$p \rightarrow \leftarrow \bar{p}$ collides

$W \rightarrow e\ \nu$ 80.2 GeV/c^2

$Z^0 \rightarrow \begin{cases} e^+ e^- \\ \mu^+ \mu^- \end{cases}$ 91.2 GeV/c^2

$p \rightarrow$ ▨ $\rightleftharpoons \bar{p}$ cooling

1989 LEP (CERN)



91.173 GeV/c^2

$E_{Lab} \sim 8 \times 10^{12}$ GeV

Z^0 $e^- e^+ \rightarrow Z^0 \rightarrow e^- e^+$

$W^\pm$ $e^- e^+ \rightarrow W^+ W^- \rightarrow e^+ e^- \nu \bar{\nu}$

160 GeV



1998年 $W^\pm$ LEP II

SSC

20 TeV × 20 TeV

LHC (CERN)

Large Hadron Collaider 7 TeV × 7 TeV

Super symmetric Particle

(SUSY)

{ 力: gauge 粒子 (spin 1)
{ 物質: quark, lepton (spin ½)

すべての粒子に boson と fermion

photino ($\tilde{\gamma}$)

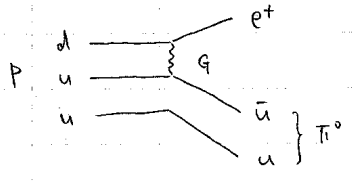
neutralino

sneutrino

grand Unified Theory

GUT

Strong - Weak - EM 統一



$\tau(p) > 8 \times 10^{32}$ yr

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4つの力

strong interaction

p-n 間 interaction $\quad \Delta t \cdot \Delta E \lesssim \hbar$

{ strong $\quad R \simeq c\Delta t \leq \frac{\hbar}{mc}$

{ short range $\quad R \sim 10^{-15}\,m \rightarrow mc^2 = \frac{\hbar c}{R} \sim 100\,MeV$

π: 140 MeV

Klein-Gordon

$$\nabla^2\psi - \frac{m^2c^2}{\hbar^2}\psi - \frac{1}{c^2}\frac{\partial^2\psi}{\partial t^2} = 0$$

{ time independent

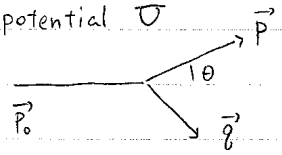
{ $\psi \rightarrow U$

$$\nabla^2 U(r) = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial U}{\partial r}\right) = \frac{m^2c^2}{\hbar^2}U(r) \quad \rightarrow \quad U(r) = \frac{g}{4\pi r}e^{-\frac{r}{R}}$$

$R = \frac{\hbar}{mc}$

Yukawa Potential による Scattering

potential U



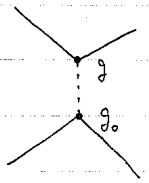
· angular defraction

mom. Transfer : $\vec{q}$

Fourier Transformation

$$f(\vec{q}) = g_0\int U(\vec{r})\,e^{i\vec{q}\cdot\vec{r}}\,d\bar{V}$$

g_0 : coupled constant



central potential $U(\vec{r}) = U(r)$

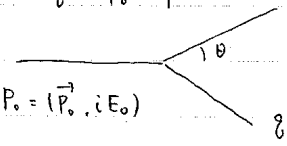
$\vec{q}\cdot\vec{r} = qr\cos\theta$

$d\bar{V} = r^2 d\phi \sin\theta\, dr \qquad \hbar = c = 1$

$$f(\vec{q}) = g_0 g\int_0^\infty e^{-mr}\cdot\frac{(e^{iqr} - e^{-iqr})}{2iq}\,dr = \frac{g_0 g}{m^2 + |\vec{q}|^2}$$

4-mom. transfer による Scatt. ampl.

$\vec{q} = \vec{P_0} - \vec{P}$



$q^2 = (P_0 - P)^2 - (E_0 - E)^2$ $\quad q^2 = \vec{q}^{\,2} - W^2$

$f(q^2) = \dfrac{g g_0}{q^2 + m^2}$

$\dfrac{d\sigma}{dq^2} = |f(q^2)|^2$

energy transfer 大きい

$m = 0$ のとき

$\dfrac{d\sigma}{dq^2} \propto 1/q^4$ Rutherford scattering

1936 Anderson $\quad 2.2 \times 10^{-6}$ sec

$\mu^- \to e^- (+ \nu_\mu + \bar{\nu}_\mu)$

cosmic ray

1947 Lattes

$\pi^- \to \mu^- + \bar{\nu}_\mu$

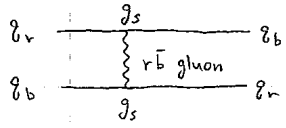
1943 坂田 2中間子論

QCD

gluon exchange

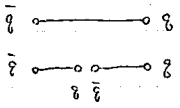
strong force の origin

6 types of strong charge



$V_s = -\dfrac{4}{3}\dfrac{\alpha_s}{r} + kr$

$\begin{cases} \text{small } r & : \text{1st term} \\ \text{large } r & : \text{2nd term} \end{cases}$



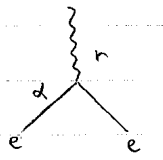
EM-int

$V_e(r) = \dfrac{e^2}{4\pi r} = \dfrac{e^2}{4\pi\hbar c} \cdot \dfrac{\hbar c}{r}$

fine structure const. $\alpha = 1/137$

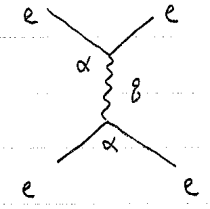
$\dfrac{\alpha_s}{\alpha} \simeq 100$

① Photo-electric Effect



$\sigma \propto \alpha$

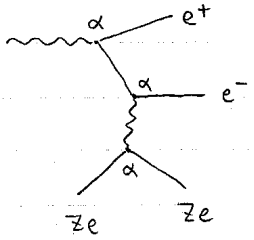
② Rutherford Scatt.



$\sigma \propto \alpha^2/q^4$

③ Pair creation





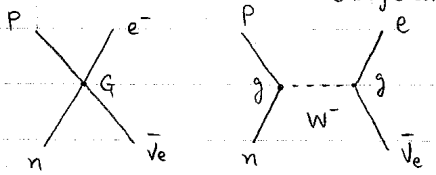
$\sigma \propto \alpha^3(z^2)$

Weak - interaction

$\begin{cases} n \to p + e^- + \bar{\nu}_e \\ \bar{\nu}_e + p \to n + e^+ \end{cases}$

Weak $\quad \Sigma^- \to n + \pi^- \quad \tau = 10^{-10}$ sec

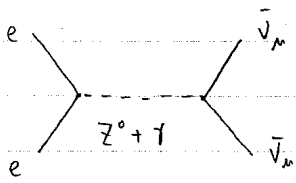
E-M $\quad \Sigma^0 \to \Lambda + \gamma \quad \tau = 10^{-19}$ sec



$\frac{\text{Weak coupling}}{\text{EM} \quad "} = \sqrt{\text{life ratt}} \simeq 10^{-5}$

$f(q^2) = \frac{g^2}{(q^2 + m^2_{W,Z})}$

$\bar{\nu}_\mu + e \to \bar{\mu}_\mu + e$



neutral current

$q^2 \ll M^2$ Fermi int.

point like

$G = \frac{g^2}{M_W^2} \sim 10^{-5}$ GeV^{-2}

β - decay

$M^2 \gg q^2$

$\frac{d\sigma}{dq^2} \sim \frac{g^2}{M^2} = G^2$

$\sigma^{tot} \sim G^2 q^2_{max}$

Unitarity ⇐ 무한대

Gravitational - Interaction

$\frac{kM^2}{r^2} \longrightarrow \frac{kM^2}{2\pi\hbar c} = 4.6 \times 10^{-40}$

Plank mass $M_p \doteq \left(\frac{4\pi\hbar c}{k}\right)^{\frac{1}{2}} = 2 \times 10^{19}$ GeV

Parity, Charge Conjugation } violated Weak Interaction

Time Reversal : no evidence of violation

CP : OK expect K^0 decay

$K_L^0 \to \pi\pi$ exist

T : It should be violated if CPT is O.K.

CPT : O.K.

Parity operator P $\quad P\psi(\vec{r}) = \psi(-\vec{r})$

$P^2 = 1 \quad P = \pm 1$ eigenvalue

scalar	$P = +1$	scalar	
	-1	pseudo scalar	
vector	$+1$	axial vector	$\vec{M} = \vec{r} \times \vec{P}$
	-1	vector	
tensor	± 1		

粒子 Intrinsic Parity

fermion ⟷ antifermion : opposit parity

boson ⟷ antiboson : same parity

proton : convention

strong, E-M } conserve

Weak : non-conserve

反応前後の parity 保存

before P_i

after P_f

Hamiltonian が空間反転で不変なら

$$\int \psi_f^*(x) H \psi_i(x) d\tau = \int \psi_f^*(-x) H \psi_i(-x) d\tau = P_i P_f \int \psi_f^*(x) H \psi_i(x) d\tau \qquad P_i = P_f$$

No. 8

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Spin 0 の粒子が 2 ヶある状態の parity

relative angular mom : ℓ

relative motion の wave fn

$f_\ell(x-y) = (-1)^\ell f_\ell(y-x)$

$[\, Y_\ell^m(\pi-\theta, \phi+\pi) = (-1)^\ell Y_\ell^m(\theta,\varphi) \,]$

$\begin{cases} P_a, P_b : a,b\text{粒子の Parity} \\ \phi_a\ \phi_b : \quad 〃 \quad \text{wave fn} \end{cases}$ $\begin{cases} \phi_a(x) = P_a \phi_a(-x) \\ \phi_b(y) = P_b \phi_b(-y) \end{cases}$

全体の CM 系 wave function

$\phi_a(x)\, f_\ell(x-y)\, \phi_b(y) \longrightarrow$ 全 Parity $P = P_a P_b (-1)^\ell$

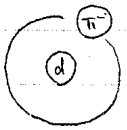
Pion Parity

π : boson (スピン $S=0$)

$\pi^- + d \longrightarrow n + n$

$(2n + 1)$

左辺



mesic X-ray の測定

S-state

$S_d = 1 \quad S_\pi = 0$

total angular mom. $J = 1$

右辺 spin $\frac{1}{2}$ の 2 ヶの粒子の spin wave function

Clebsh-Gorden Coefficient

粒子の入れ換えについて symmetric → triplet ($S=1$)

$\begin{cases} \alpha(1,1) = \psi_1(\frac{1}{2},\frac{1}{2})\,\psi_2(\frac{1}{2},\frac{1}{2}) \\ \alpha(1,0) = \frac{1}{\sqrt{2}}\left[\psi_1(\frac{1}{2},\frac{1}{2})\,\psi_2(\frac{1}{2},-\frac{1}{2}) + \psi_2(\frac{1}{2},\frac{1}{2})\,\psi_1(\frac{1}{2},-\frac{1}{2})\right] \\ \alpha(1,-1) = \end{cases}$

anti-symmetric → singlet ($S=0$) $\quad \alpha(0,0) =$

$\begin{cases} \text{spin wave fn} & (-1)^{S+1} \\ \text{orbit} & (-1)^L \end{cases}$

overall $(-1)^{S+L+1} = -1$ $\qquad L+S$ even $\qquad L=S=1 \quad {}^3P_1$

parity $= (-1)^L (P_n)^2 = (-1)^L = -1$

n.p +1 parity

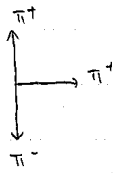
$\begin{cases} d & \text{parity } +1 \\ \pi & 〃 \quad -1 \end{cases}$ $\qquad \pi$ meson pseudo scalar $\quad 0^{-1}$

K^+ - decay $\begin{cases} \to \pi^+ \pi^0 \\ \to \pi^+ \pi^- \pi^+ \end{cases}$

π-π 系 ang. mom. : 0

" parity : +1

$\begin{cases} P_\pi : -1 \\ S_{\pi,K} : 0 \end{cases}$



π-π-π 系

$\begin{cases} \pi^+\pi^- \text{ と } \pi^+ \text{ の CM系 ang. mom. } 0 \\ \text{ang. mom による parity } +1 \end{cases}$

3π parity $= (-1)^3 = -1$

全 parity $= -1$ τ-θ puzzle

Lee . Yang Proposal

Weak Interaction で parity 保存せず

β - decay

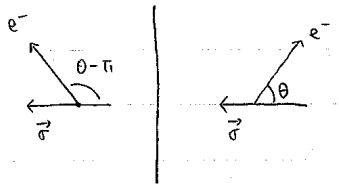
$n \to p + e^- + \bar{\nu}_e$

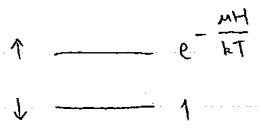
Wu (呉健雄)

$^{60}Co \to {}^{60}Ni + e^- + \bar{\nu}_e$

5+ 4+

$\Delta J = 1$ G-T ガモフ テーラー





$\vec{J}$, $\vec{P}_e$ } 角相関

e : θ の方向に出る確率

$I(\theta)\sin\theta\, d\theta = \{F_1 + F_2(\vec{J}\cdot\vec{P}_e)\}\sin\theta\, d\theta$

$= \{F_1 + F_2 J \cdot P_e \cos\theta\}\sin\theta\, d\theta$

空間反転

$I(\pi-\theta)\sin\theta\, d\theta = \{F_1 - F_2 \cdot J \cdot P_e \cos\theta\}\sin\theta\, d\theta$ $\quad I(\theta) = I(\pi-\theta)$

$F_2 = 0$

Exp $F_2 \neq 0$

Lederman

$\mu^+ \to e^+ + \bar{\nu}$

$\vec{\sigma}_\mu$ $\vec{P}_{e^+}$

$\pi^+ \to \mu^+ + \nu$

1/2 -1/2

↑ helicity -1

helicity $H = \dfrac{\vec{\sigma}\cdot\vec{P}}{|\vec{P}|}$: +1, 0, -1

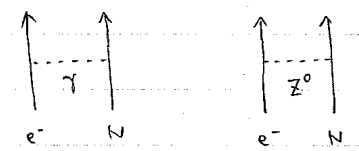
μ^+ : 100 % pol

left scalar right handed

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Parity-violation in ep scattering



(E-M 沒有 Parity-viol 項)

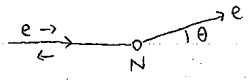
Scattering amplitude

$$f = f_{EM} + f_{WK}$$

cross section

$$\sigma_{eN} \propto f^2 = |f_{EM}|^2 \cdot \left\{ 1 + \frac{f_{WK} f_{EM}^* + f_{WK}^* f_{EM}}{|f_{EM}|^2} + \frac{|f_{WK}|^2}{|f_{EM}|^2} \right\}$$

Opposit helicity (+,−) 之間的干涉



$$A_L = \frac{\sigma_+ - \sigma_-}{\sigma_+ + \sigma_-} = \frac{|f_+|^2 - |f_-|^2}{|f_+|^2 + |f_-|^2} \sim \frac{f_{WK} f_{EM}^* + f_{WK}^* f_{EM}}{|f_{EM}|^2}$$

$$\begin{cases} f_{EM} \simeq 4\pi\alpha / q^2 \\ f_{WK} \approx G_F = \dfrac{g^2}{M_Z^2} \end{cases}$$

$$\simeq \frac{G_F q^2}{4\pi\alpha} \simeq \frac{q^2}{M_Z^2} \simeq 10^{-4}$$

SLAC Prescott
Tayler

$$\begin{cases} g = e & |q| \sim 1\,\text{GeV} \\ \alpha = 1/137 \end{cases}$$

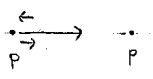
Linac 16 ~ 22 GeV/c
e^- on d

$$\frac{A_L}{q^2} = -(9.5 \pm 1.6) \times 10^{-5}\ (\text{GeV}/c)^{-2}$$

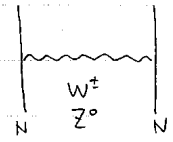
$\sin\theta_W = 0.22$ winberg-angle

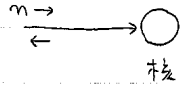
Parity violetion in NN Interaction

Yuan Los Alamos

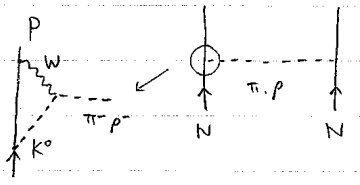


$$A_L = (2.4 \pm 1.1) \times 10^{-7}$$





$A_L \sim 10^{-1}$



n

$$\frac{V_{pnc}}{V_0} \sim \frac{G_F}{\alpha_s} \sim G_F m_\pi^2 \sim 2 \times 10^{-7}$$

↖ Strong $\frac{1}{m_\pi^2}$

Charge conjugation

- 電荷の sign, magnetic moment の sign を変える
- 古典 Maxwell eq は invariant Charge current density E. H の sign をかえても invariant
- Relativestic quantum mechanics

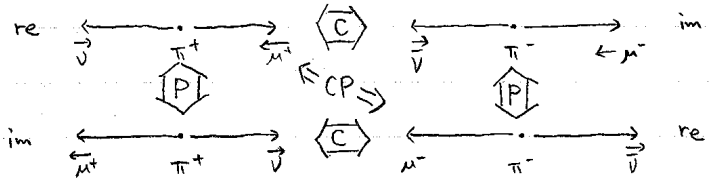
particle ⟷ antiparticle

	particle	antiparticle
mass	M	M
charge	$\mathcal{E}$	$-\mathcal{E}$
mag. mom	μ	$-\mu$
baryon #	B	$-B$
lepton #	L	$-L$
spin	S	S

strong, EM int. conserve

$A+B \to C+D \qquad \bar{C}+B \to \bar{A}+D$

Weak interaction



CP

$K^+ + n \to K^\circ + P$
$(\bar{s}d)$

$K^- + P \to \bar{K}^\circ + n$
$(s\bar{d})$

C $\begin{cases} K^\circ \to \bar{K}^\circ \\ \bar{K}^\circ \to K^\circ \end{cases}$
P $\begin{cases} K^\circ \to -K^\circ \\ \bar{K}^\circ \to -\bar{K}^\circ \end{cases}$

CP $\begin{cases} K^\circ \to -\bar{K}^\circ \\ \bar{K}^\circ \to -K^\circ \end{cases}$

$\begin{cases} K^\circ + \bar{K}^\circ \to -(K^\circ + \bar{K}^\circ) \\ K^\circ - \bar{K}^\circ \to (K^\circ - \bar{K}^\circ) \end{cases}$

P $\begin{cases} \pi\pi & P=1 \\ \pi\pi\pi & P=-1 \end{cases}$

C $\begin{cases} \pi^+ \to \pi^- \\ \pi^\circ \to \pi^\circ \\ \pi^- \to \pi^+ \end{cases}$

CP $\begin{cases} \pi\pi & CP=1 \\ \pi\pi\pi & CP=-1 \end{cases}$

$|K_1^\circ\rangle = \frac{1}{\sqrt{2}}(|K^\circ\rangle + |\bar{K}^\circ\rangle)$ $\quad CP|K_1^\circ\rangle = -|K_1^\circ\rangle \to 3\pi$ CP不変なら $= K_L^\circ$ $\quad \tau_L = 5.2\times10^{-8}$ s

$|K_2^\circ\rangle = \frac{1}{\sqrt{2}}(|K^\circ\rangle - |\bar{K}^\circ\rangle)$ $\quad CP|K_2^\circ\rangle = +|K_2^\circ\rangle \to 2\pi$ 〃 $= K_S^\circ$ $\quad \tau_S = 0.89\times10^{-10}$ s

$|K^\circ\rangle$, $|\bar{K}^\circ\rangle$: states of definite strangeness

$|K_1^\circ\rangle$, $|K_2^\circ\rangle$: 〃 CP

$|K_S\rangle$, $|K_L\rangle$: 〃 life

まじり合う
$|K^\circ\rangle \longleftrightarrow |\bar{K}^\circ\rangle$

はじめ K° だけでも時間がたつと $\bar{K}^\circ$ に変る

K_1° K_2° = decay mode が異なるので時間がたつと

K_2° は decay

K_1° は残る

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$|K^0\rangle = \frac{1}{\sqrt{2}}(|K_1^0\rangle + |K_2^0\rangle)$　　$t=0$ で K^0 100% としたときの t sec 後の状態 $\psi(t)$ を求める

$|\bar{K}^0\rangle = \frac{1}{\sqrt{2}}(|K_1^0\rangle - |K_2^0\rangle)$　　準安定準位の とり あつかい　$\sim \exp\left[-\frac{i}{\hbar}E_1 - \frac{\Gamma_1}{2\hbar}\right]t$

$$\psi(t) = \frac{1}{\sqrt{2}}\left\{|K_1^0\rangle \exp(-im_1 t)\exp\left(-\frac{\Gamma_1 t}{2}\right) + |K_2^0\rangle \exp(-im_2 t)\exp\left(-\frac{\Gamma_2 t}{2}\right)\right\}$$

$\begin{cases} m_1, m_2 : K_1^0, K_2^0 \text{ の質量} \\ \Gamma_1, \Gamma_2 : \quad " \quad \text{の decay rates } (1/\tau_1, 1/\tau_2) \end{cases}$

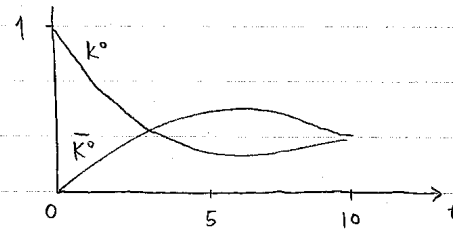
$$= \frac{1}{2}\left[|K^0\rangle\left\{\exp(-im_1 t)\exp\left(-\frac{\Gamma_1 t}{2}\right) + \exp(-im_2 t)\exp\left(-\frac{\Gamma_2 t}{2}\right)\right\}\right.$$
$$\left. + |\bar{K}^0\rangle\left\{ \quad " \quad - \quad " \quad \right\}\right]$$

t の時 K^0 検出確率

$$|\langle K^0|\psi(t)\rangle|^2 = \frac{1}{4}\left\{\exp(-\Gamma_1 t) + \exp(-\Gamma_2 t) + 2\exp\left(-\frac{\Gamma t}{2}\right)\cos \delta m \cdot t\right\}$$

$\Gamma = \Gamma_1 + \Gamma_2$　　$\delta m = m_1 - m_2$

$\delta m \cdot \tau_1 = 0.5$　　$\left(\tau_1 = \frac{\hbar}{\Gamma_1}\right)$



$\bar{K}^0$ の量を K^0 source からの距離 の function として測定

$\delta m \cdot \tau_1 = 0.477$

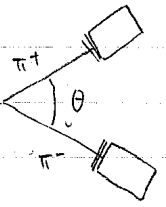
$\delta m = 3.52 \times 10^{-6}$ eV

$m_2 > m_1$

$\frac{\delta m}{m} = 0.7 \times 10^{-14}$

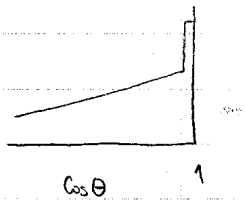
1964 Fitch et al

$K_L \to \pi^+\pi^-$　　$K_L^0 \to \begin{cases} \pi^+\pi^-\pi^0 \\ \pi^+\pi^- \ 0.25\% \end{cases}$



CP - violation

$$\beta = \frac{(K_L^0 \to \pi^+\pi^-)}{(K_L^0 \to \text{charged total})} \sim 2.5 \times 10^{-3}$$



$$|\eta_{+-}| = \frac{\text{Amp}(K_L \to \pi^+\pi^-)}{" \ (K_S \to \pi^+\pi^-)} = \varepsilon + \varepsilon' \sim 2.3 \times 10^{-3}$$

$$|\eta_{00}| = \frac{\text{Amp}(K_L \to \pi^0\pi^0)}{" \ (K_S \to \pi^0\pi^0)} = \varepsilon - 2\varepsilon' \sim 2.3 \times 10^{-3}$$

$\begin{cases} \text{super weak} & \varepsilon' = 0 \\ 6\text{-quark} & \varepsilon' \neq 0 \end{cases}$

$$\varepsilon' = \frac{i}{2}\frac{\text{Im}\,A_2}{A_0} e^{i(\delta_2 - \delta_0)}$$

$\begin{cases} A_0 : I = 0 \text{ の } 2\pi \text{ に decay する amplitude} \\ A_2 : I = 2 \quad " \end{cases}$

$\begin{cases} \delta_0 : I = 0 \text{ の } 2\pi \text{ の final state int による phase shift} \\ \delta_2 : I = 2 \quad " \end{cases}$

CPT : invariant in quantum field theory

CP : violate $\longrightarrow$ T : violate

Super weak model

Wolfenstein 1964

$H_{wk} = \alpha H'_{wk} + \beta H'_{s.wk}$

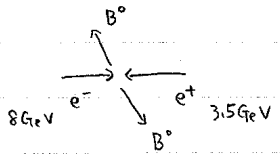
H'_{wk} : normal weak interaction $\Delta S = 1$ CP conserve

$H'_{s.wk}$: $\Delta S = 2$ CP violating

$\varepsilon \simeq \frac{\beta}{\alpha} \times 10^5 \rightarrow \frac{\beta}{\alpha} \sim 10^{-7}$

B-meson decay

B-factory KEK, Stanford



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1964 Ω粒子の発見

1964 Gell-man, Zweig

{ baryon # 1/3
{ charge (2/3)e (1/3)e

基本粒子とその反粒子

	Q	I	I_3	S
u	2/3	1/2	1/2	0
d	-1/3	1/2	-1/2	0
s	-1/3	0	0	-1
$\bar{u}$	-2/3	1/2	-1/2	0
$\bar{d}$	1/3	1/2	1/2	0
$\bar{s}$	1/3	0	0	-1

quark

{ meson $q\bar{q}$
{ baryon qqq
{ anti-baryon $\bar{q}\bar{q}\bar{q}$

{ $q = (u, d, s)$ SU(3) 3重項
{ $\bar{q} = (\bar{u}, \bar{d}, \bar{s})$ 〃 反3重項

{ $\pi^- \sim d\bar{u}$
{ $K^- \sim s\bar{u}$

$q\bar{q}$ state

$3 \otimes 3^* = 8 \oplus 1$

{ $p \sim uud$
{ $n \sim udd$
{ $\Omega^- \sim sss$

qqq

$3 \otimes 3 \otimes 3 = 10 \oplus 8 \oplus 8 \oplus 1$

{ 10重項 spin 3/2 baryon 完全対称
{ 8重項 spin 1/2 baryon 一意的でない
{ 1重項 spin 1/2

Spin 8重項

$p \sim (udu + duu - 2uud)/\sqrt{6}$

$n \sim (dud + udd - 2ddu)/\sqrt{6}$

⋮

1重項

$\Lambda^{*0} \sim (uds - usd + dsu - dus + sud - sdu)/\sqrt{6}$

他の例

$p \sim (udu - duu)/\sqrt{2}$

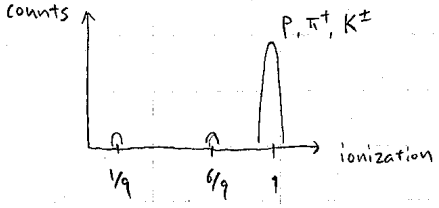
$n \sim (dud - udd)/\sqrt{2}$

Free quark search

All searches since 1977 have had negative results.

{ · Accelerator { hadron - hadron
{ { $e^+ - e^-$
{ { $e^- -$ hadron
{ · Cosmic Ray
{ · 物質中に存在する quark

Ionization $\propto z^2$



pp $\bar{p}p$ reactions $< 10^{-11}$ for q/π

ep $< 10^{-5}$ for q/ interaction

e^+e^- $< 10^{-3}$ for $\frac{q + X}{e^+e^- \to \mu^+\mu^-}$

cosmic ray $< 10^{-10}$ for primary nuclear

Pre-existing-quark 探索

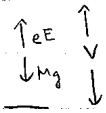
fractional charge $< 10^{-24}$ q/ sea water

moon rocks

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millikan



$eE = Mg$

1/3 e 1回だけ

1977 : Fairbank

oscillating electric field

magnetic field の中に うかす3

100 μg Nb 球

$\left.\begin{matrix}(-0.34 \pm 0.1)e \\ (+0.33 \pm 0.11)e\end{matrix}\right\}$ 5.7

SU(6) model

1964 : B. Sakita, Gersey, Rodicati

quark 間の力は spin independent

meson spin 0 or 1

$\frac{1}{2} \otimes \frac{1}{2} \to 0 \oplus 1$

meson parity $P = P_q P_{\bar{q}} (-1)^l$

$J^P = 0^-, 1^-$

(u. d. s) それぞれ ↑↓

SU(6) 変換に対し不変

$q\bar{q}$ s-wave

$6 \otimes 6^* = 35 \oplus 1$

$35 = 1 \times 8 + 3 \times 1 + 3 \times 8$

$\begin{cases} 0^- & 8\text{重項} \\ 1^- & 1\text{重項} \times 3 \\ 1^- & 8\text{重項} \times 3 \end{cases}$

$\begin{cases} (u\bar{u} + d\bar{d} - 2s\bar{s})/\sqrt{6} & 8\text{重項の1つ} \\ (u\bar{u} + d\bar{d} + s\bar{s})/\sqrt{3} & 1\text{重項} \end{cases}$

$\begin{cases} \omega^0 \sim (u\bar{u} + d\bar{d})/\sqrt{2} \\ \phi^0 \sim s\bar{s} \end{cases}$

Magnetic moment of Baryons

SU(6)

Dirac 粒子の magnetic moment

$$\vec{\mu}_i = \frac{e_i \hbar}{2m_i c} \vec{\sigma}_i$$

$l = 0$ とする

baryon mag-mom は

q の mag-mom の vector 和

$\begin{cases} \mu_\Lambda = \mu_s \\ \mu_\Sigma = \frac{4}{3}\mu_u - \frac{1}{3}\mu_s \\ \vdots \\ \Omega^- = 3\mu_s \end{cases}$

	theory	exp.
Ω^-	-1.84	-1.94 ± 0.22

proton uud

uu : triplet state

$\chi_1 (J=1)$

d :

$\chi_2 (J=\frac{1}{2})$

C-G $\psi(\frac{1}{2}, \frac{1}{2}) = \sqrt{\frac{2}{3}} \underbrace{\chi_1(1,1)}_{2\mu_u} \underbrace{\chi_2(\frac{1}{2}, -\frac{1}{2})}_{-\mu_d} - \sqrt{\frac{1}{3}} \underbrace{\chi_1(1,0)}_{0} \underbrace{\chi_2(\frac{1}{2}, \frac{1}{2})}_{+\mu_d}$

$\mu_p = \frac{2}{3}(2\mu_u - \mu_d) - \frac{1}{3}\mu_d = \frac{4}{3}\mu_u - \frac{1}{3}\mu_d$

neutron

$\mu_d \leftrightarrow \mu_u$

$\mu_n = \frac{4}{3}\mu_d - \frac{1}{3}\mu_u$ $\quad m_u = m_d \to \frac{\mu_n}{\mu_p} = -\frac{2}{3}$

$\sim 330 \text{MeV}/c^2$ (exp.) $= -0.685$

mag-mom の absolute value 再現可能

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charm particle

$\bar{c}c$: charmonium

$D^+(c\bar{d}) \to \bar{K}^0\pi^+,\ K^-\pi^+\pi^-$

$D^0(c\bar{u}) \to K^-\pi^+,\ \bar{K}^0\pi^+\pi^-$

$D_s^+ \leftarrow F^+(c\bar{s}) \to \eta^0\pi^+,\ K^+\bar{K}^0$

$D^-(d\bar{c}) \to \cdots$

$\bar{D}^0(\bar{c}u) \to \cdots$

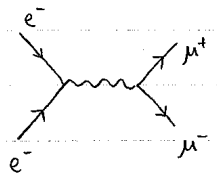
$D_s^- \leftarrow F^-(s\bar{c}) \to \cdots$

$e^+e^- \to \mu^+\mu^-$

$\sigma(e^+e^- \to \gamma \to \mu^+\mu^-) \to \dfrac{4\pi\alpha^2\hbar c^2}{3s}$

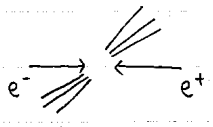
Dirac eq

s: 全エネルギーの自乗



$e^+e^- \to \gamma \to q\bar{q} \to$ hadrons

↑ 2方向の jets



$\dfrac{d\sigma}{d\cos\theta} \propto 1+\cos^2\theta$

q: spin $\frac{1}{2}$

$\sigma(e^+e^- \to u\bar{u}) = \left(\frac{2}{3}\right)^2\left(\frac{4\pi\alpha^2\hbar c^2}{3s}\right)$

$\sigma(\ " \to d\bar{d}) = \left(-\frac{1}{3}\right)^2(\ \ " \ \ )$

$\sigma(\ " \to s\bar{s}) = (\ " \ )(\ " \ )$

} color 自由度 ×3

$R = \dfrac{\sigma_{had}}{\sigma_{\mu\mu}} = \dfrac{e_u^2+e_d^2+e_s^2}{e^2} = \dfrac{2}{3}$

Exp) $R = \begin{cases} 2 & \sqrt{s} \le 4\,\text{GeV} \\ 5 & \gtrsim 4\,\text{GeV} \end{cases}$

Theory: charm をのぞくと $R = \frac{10}{9}\times 3 = \frac{10}{3}$

τ 粒子の effect

↓

$R \sim 4.5$

t があれば $R = \frac{11}{3} + \frac{4}{3} = 5$

Lederman et al at Fermilab

$p + Be \to \mu^+\mu^- + X$

$M_\Upsilon = 9.46\ \text{GeV}/c^2$

$\Upsilon : b\bar{b}$

1 より上の energy では $R = \frac{11}{3}$ のはず

$3 \times \dfrac{e_u^2+e_d^2+e_s^2+e_c^2+e_b^2}{e^2} = \dfrac{11}{3}$

e^2 : $4/9$ $1/9$ $1/9$ $4/9$ $1/9$

Tristan (Tsukuba)

top quark search

LEP (CERN)

e^+e^- 2×45 GeV

Z^0, $W^\pm$

↑ 2×90 GeV

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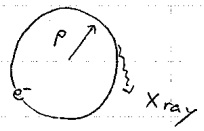
$Z^0 \rightarrow$ { hadrons (u, d, s, c, b) / e^+e^-, $\mu^+\mu^-$, $\tau^+\tau^-$ / $\nu\bar{\nu}$

$t \rightarrow W^+ + b$
→ $e^+\nu$ or $\mu^+\nu$
→ $u\bar{d}$ or $c\bar{s}$

$\Gamma_{invisible}(\nu\bar{\nu}) = \Gamma_{Z^0} - \Gamma_{had} - \Gamma_{e^+e^-} - \Gamma_{\mu^+\mu^-} - \Gamma_{\tau^+\tau^-}$

$N_\nu = \dfrac{\Gamma_{invisible}}{\Gamma_\nu^{SM}}$ $\Gamma_\nu^{SM} = 0.167\ GeV$

Standard model



$\langle N_\nu \rangle = 2.97 \pm 0.05$

1994 Fermi lab

$p + \bar{p} \rightarrow t\bar{t} + X$

uud $\bar{u}\bar{u}\bar{d}$



0.01 μm の精度でぶつける

$t\bar{t} \rightarrow$ { W^+W^- → { $ee\nu\bar{\nu}$ or $e\mu\nu\bar{\nu}$ or $\mu\bar{\mu}\nu\bar{\nu}$ + 2 jet } 2ケ / + 2 jet → $e\nu$ or $\mu\nu$ + 4 jets } 4ケ

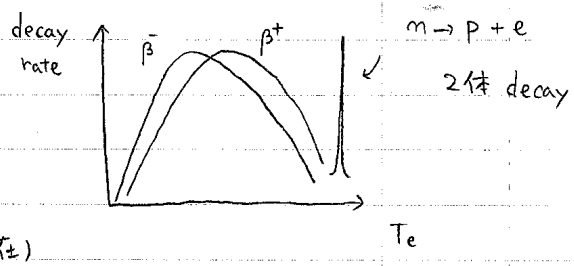
t の mass $174 \pm 10^{+13}_{-12}\ GeV/c^2$

β - decay

1914 Chadwick β-線 連続スペクトル

1930 Pauli (ETH) $n \rightarrow p + e^- + \nu$ ν_2

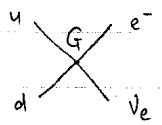
spin $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ $\bar{\nu}_e$ (現在)



$n \rightarrow p + e$

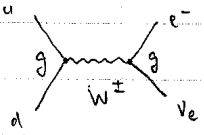
2体 decay

$d \rightarrow u + e^- + \bar{\nu}_e$



universal Fermi Interaction

$g^2 \ll M_W^2$



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2 Type of β-decay

Fermi 型 $\Delta\vec{J} = 0$

$^{14}O \rightarrow {}^{14}N^* + e^+ + \nu$

J^P 0^+ 0^+

Gamov-Teller 型 $\Delta\vec{J} = 0$

$^6He \rightarrow {}^6Li + e^- + \bar{\nu}$

J^P 0^+ 1^+

(Pure G-T)

mix 型

$n \rightarrow p + e^- + \bar{\nu}$

J^P $\frac{1}{2}^+$ $\frac{1}{2}^+$

$J \rightarrow J$ $(J \neq 0)$

Pure Fermi

Fermi ($\Delta\vec{J} = 0$) G-T ($\Delta\vec{J} = 1$)

$n \rightarrow p + e^- + \bar{\nu}_e$ $n \rightarrow p + e^- + \bar{\nu}_e$

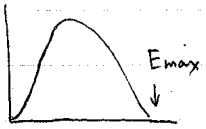
↑ ↑ ↓ ↑ ↑ ↓ ↑ ↑

Fermi

neutron decay rate

$$W = \frac{2\pi}{\hbar} \sum_f |H|^2 \delta(E_f - E_i) \left(\frac{dN}{dE_0}\right)$$

E_0 : final state energy



$(E_0) = E + E_\nu = E_{max}$

↑ electron total energy

$M_n c^2 - M_p c^2 \sim 1.3\,MeV$ T_p : neglect

$H = G \int \psi_p^*(x)\, \psi_e^*(x)\, \psi_\nu^*(x)\, \psi_n(x)\, d\tau$ G : coupling const

$U(x - x') \longrightarrow G\,\delta(x - x')$

β-decay の interaction の E_ν : 0

ψ を plane wave で近似 $\frac{e^{-i\vec{p}\cdot\vec{x}/\hbar}}{\sqrt{V}}$

$$\frac{dN}{dE_0} = \frac{V^2 p_e^2 p_\nu^2 dp_0\, dp_\nu\, d\Omega_e d\Omega_\nu}{(2\pi\hbar)^2}$$

$H = \frac{G}{V}\,\delta_{\vec{p}_p + \vec{p}_e + \vec{p}_\nu,\, \vec{p}_n}$ proton は独立でない

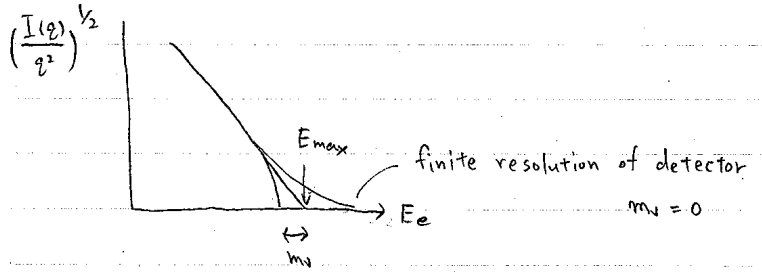
$$W = \frac{2\pi V^2}{\hbar (2\pi\hbar)^6} \int p_\nu^2\, dp_\nu\, p_e^2\, dp_e\, d\Omega_\nu\, d\Omega_e\, \delta(E_\nu + E - E_{max})\, \frac{G^2}{V^2}$$

$$= \frac{G^2}{2\pi^3 \hbar^7 c^3} \int_0^{p_{max}} (E_{max} - E)^2\, dp_e^2\, dp_e = \int_0^{p_{max}} I(p)\, dp \qquad p_\nu = E_\nu / c$$

$$I(p) = \frac{G^2}{2\pi^3 \hbar^7 c^3}\, p^2 \left[E_{max} - \underbrace{\sqrt{(Mc^2)^2 + (pc)^2}}_{E} \right]^2 \qquad \left(\frac{I(p)}{p^2}\right)^{1/2} \propto (E_{max} - E)$$

No.

Date 94 · 12 · 16



Kurie plot $m_\nu < 7.3\ eV$